Magnitude scale: (For this part look up Gregory and Zellick book, Chap 11, P226 onward)

Quantitative measurement of brightness (i.e., intensity) requires the definition of magnitude.

Hipparchus was the first person to catalog the stars based on their *apparent* magnitude. He qualitatively assigned m=1 to the brightest star and dimmest star visible to naked eye m=6. Since his time, astronomers have extended and refined his apparent magnitude scale.

Higher *apparent* magnitude → Lower brightness.

The "brightness" of a star is quantitatively measured in terms of the radiant flux F (total energy per unit time per unit area) received from the star.

Human eye responds to the difference in the logarithms of the brightness of two luminous objects. This theory led to a scale in which a difference of 1 magnitude between two stars implies a constant ratio between their brightnesses.

By the modern definition, a difference of 5 magnitudes corresponds exactly to a factor of 100 in brightness, so a difference of 1 magnitude corresponds exactly to a brightness ratio of $100^{1/5} \simeq 2.512$. Thus a first-magnitude star appears 2.512 times brighter than a second-magnitude star, $2.512^2 = 6.310$ times brighter than a third-magnitude star, and 100 times brighter than a sixth-magnitude star.

Hipparchus's scale has been extended in both directions, from $m = -26.83$ for the Sun to approximately $m = 30$ for the faintest object detectable.

The radiant flux F can be written as $F = \frac{L}{4\pi r^2}$ where L is total luminosity (total emitted power) of the star (inverse square law for light).

By measuring the radiant flux (or the luminosity) and using the fact that a difference of 5 magnitudes corresponds exactly to a factor of 100 in brightness, we can derive a quantitative formula for *apparent* magnitude.

If two stars have apparent brightnesses F_1 and F_2 ($< F_1$) and their magnitude classes are m_1 and m_2($>m_1$), then

$$\frac{F_2}{F_1} = 100^{\frac{m_1 - m_2}{5}}$$

Taking $\log_{10}$ on both sides we get,

$$m_1 - m_2 = -2.5 \log_{10}\left(\frac{F_1}{F_2}\right)$$

An example: The apparent magnitude of the variable star ranges from 7.1 (brighter state, m_1) to 7.8 (fainter state, m_2). To find relative increase in brightness, we note:

$$2.5 \log_{10}\left(\frac{F_1}{F_2}\right) = 0.7 \Rightarrow \frac{F_1}{F_2} = 1.91$$

So, a brightness increase by factor of 2!

Task: Read about UBV (*Ultraviolet–Blue–Visual*) system in this context from ARC book.

The *absolute magnitude M* of a celestial object is defined as the magnitude it would have if it was placed at a distance of 10 pc.

Consider a star located at distance d has an *apparent* magnitude m and radiant flux F. If the star is broght to a distance of 10 pc, we can write

$$\frac{F}{F_{10}} = \frac{L/4\pi d^2}{L/4\pi(10\,pc)^2} = \frac{(10\,pc)^2}{d^2},$$

where, F_{10} is the flux that would be received if the star were at a distance of 10 pc. Now using above relationship,

$$m - M = -2.5\log_{10}\left(\frac{(10\,pc)^2}{d^2}\right) = 5\log_{10}\left(\frac{d}{10\text{ pc}}\right)$$